

MA 301
Test 3
Spring - 2015

Problem	points	out of
1		6
2		6
3		3
Score		15

Name: SOLUTIONS

HR: _____

Signature: _____

Formula Sheet	
Fourier Series:	$f(x) = \sum_{n=-\infty}^{\infty} d_n e^{i\pi n x/L} \text{ where } d_n = \frac{1}{2L} \int_{-L}^L f(x) e^{-i\pi n x/L} dx$
if $f(x)$ is even then	$d_n = \frac{1}{L} \int_0^L f(x) \cos\left(\frac{\pi n x}{L}\right) dx$
if $f(x)$ is odd then	$d_n = \frac{-i}{L} \int_0^L f(x) \sin\left(\frac{\pi n x}{L}\right) dx$

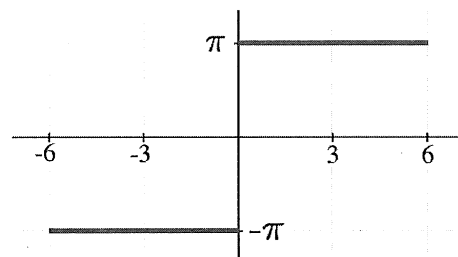
Problem 1: For the function depicted below, determine the Fourier Series in:

(a) Exponential Form

ODD $\rightarrow$

(b) Cosine/Sine Form

$L = 6$



(assume periodic extension along the x-axis)

$$d_0 = 0$$

$$\begin{aligned} d_n &= -\frac{i}{6} \int_0^6 \pi \sin\left(\frac{\pi n x}{6}\right) dx \\ &= -\frac{i\pi}{6} \left[-\frac{6}{\pi n} \cos\left(\frac{\pi n x}{6}\right) \right]_0^6 \\ &= \frac{i}{n} [\cos(\pi n) - \cos(0)] \\ &= \frac{i}{n} [(-1)^n - 1] = \begin{cases} 0 & n \text{ even} \\ -\frac{2i}{n} & n \text{ odd} \end{cases} \end{aligned}$$

Input answers here:

(a) Exponential Form: $f(x) = \sum_{n=-\infty}^{\infty} d_n e^{\frac{i\pi n x}{6}}$ where $d_n = \begin{cases} 0 & n = 0 \\ 0 & n \text{ even} \\ -\frac{2i}{n} & n \text{ odd} \end{cases}$

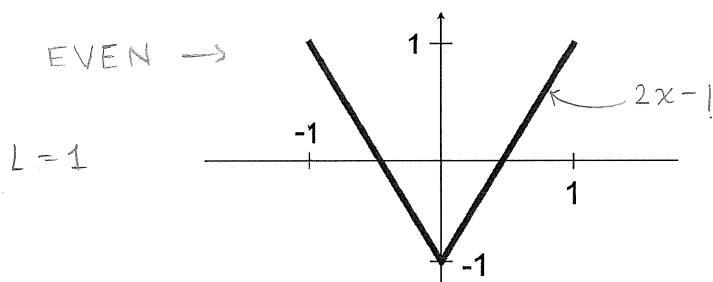
OR $= \sum_{n \text{ odd}} -\frac{2i}{n} e^{\frac{i\pi n x}{6}}$

(b) Cosine/Sine Form: $f(x) = \sum_{n=1}^{\infty} 2i d_n \sin\left(\frac{\pi n x}{6}\right)$ OR $\sum_{\substack{n=1 \\ \text{odd}}}^{\infty} \frac{4}{n} \sin\left(\frac{\pi n x}{6}\right)$

Problem 2: For the function depicted below, determine the Fourier Series in:

(a) Exponential Form

(b) Cosine/Sine Form



(assume periodic extension along the x-axis)

$$d_0 = \frac{1}{1} \int_0^1 f(x) dx = 0$$

$$d_n = \frac{1}{1} \int_0^1 (2x-1) \cos(\pi n x) dx$$

$$= \frac{2x-1}{\pi n} \sin(\pi n x) - \frac{2}{\pi n} \int \sin(\pi n x) dx$$

$$= \frac{2x-1}{\pi n} \sin(\pi n x) + \frac{2}{\pi^2 n^2} \cos(\pi n x) \Big|_0^1$$

$$= \left(\frac{1}{\pi n} \sin(\pi n) + \frac{2}{\pi^2 n^2} \cos(\pi n) \right) - \left(-\frac{1}{\pi n} \sin(0) + \frac{2}{\pi^2 n^2} \cos(0) \right)$$

$$= \frac{2}{\pi^2 n^2} (-1)^n - \frac{2}{\pi^2 n^2} = \frac{2}{\pi^2 n^2} ((-1)^n - 1) = \begin{cases} 0 & n \text{ even} \\ -\frac{4}{\pi^2 n^2} & n \text{ odd} \end{cases}$$

Input answers here:

(a) Exponential Form: $f(x) = \sum_{n=-\infty}^{\infty} d_n e^{i\pi n x}$ where $d_n = \begin{cases} 0 & n = 0 \\ 0 & n \text{ even} \\ -\frac{4}{\pi^2 n^2} & n \text{ odd} \end{cases}$

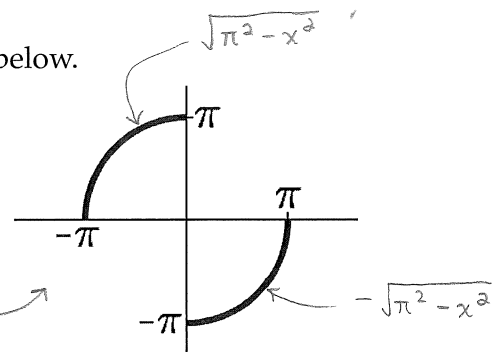
OR $= \sum_{\text{odd}} -\frac{4}{\pi^2 n^2} e^{i\pi n x}$

(b) Cosine/Sine Form: $f(x) = \sum_{n=1}^{\infty} 2 d_n \cos(\pi n x)$ or $\sum_{\substack{n=1 \\ (\text{odd})}}^{\infty} -\frac{8}{\pi^2 n^2} \cos(\pi n x)$

Problem 3: Compute d_0 and set-up, do NOT compute, d_n for the function below.

When setting up d_n , substitute in ALL known values and simplify completely, but DO NOT integrate.

(Hint: The equation of the full circle in the figure is $x^2 + y^2 = \pi^2$)



(assume periodic extension along the x-axis)

$$d_0 = 0$$

$$d_n = \frac{-i}{\pi} \int_0^{\pi} -\sqrt{\pi^2 - x^2} \sin(nx) dx = \boxed{\frac{i}{\pi} \int_0^{\pi} \sqrt{\pi^2 - x^2} \sin(nx) dx}$$

OR

$$d_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx = \boxed{\frac{1}{2\pi} \int_{-\pi}^0 \sqrt{\pi^2 - x^2} e^{-inx} dx - \frac{1}{2\pi} \int_0^{\pi} \sqrt{\pi^2 - x^2} e^{-inx} dx}$$

Input answers here:

$$d_0 = 0$$

$$d_n = \frac{i}{\pi} \int_0^{\pi} \sqrt{\pi^2 - x^2} \sin(nx) dx \quad \text{OR} \quad \frac{1}{2\pi} \int_{-\pi}^0 \sqrt{\pi^2 - x^2} e^{-inx} dx - \frac{1}{2\pi} \int_0^{\pi} \sqrt{\pi^2 - x^2} e^{-inx} dx$$